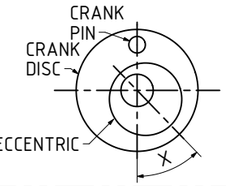
QTY.	PART NUMBER
1	09E-39-00-1-01-BASE PLATE
1	09E-39-00-1-02-ROCKING BEAM SUPPORT+SHAFT
1	09E-39-00-1-03-CYLINDER
1	09E-39-00-1-03-CYLINDER
2	09E-39-00-1-04-CROSSHEAD GUIDE COLUMN
2	09E-39-00-1-05-VALVE CHEST COVER PLATE
3	09E-39-00-1-06-EXHAUST GLAND BLOCK
2	09E-39-00-1-07-PLATFORM REAR SUPPORT COLUMN
1	09E-39-00-1-08-PLATFORM FRONT SUPPORT COLUMN
1	09E-39-00-1-09-CRANK SHAFT BEARING PLATFORM
2	09E-39-00-1-10-MAIN BEARING BLOCK
1	09E-39-00-1-11-STEAM INLET PIPE-1
1	09E-39-00-1-12-STEAM INLET VALVE
1	09E-39-00-2-01-FLYWHEEL+CRANKSHAFT
2	09E-39-00-2-02-ECCENTRIC
2	09E-39-00-2-03-ROCKING SIDE BEAM
2	09E-39-00-2-04-CROSSHEAD
2	09E-39-00-2-05-PISTON + ROD
4	09E-39-00-2-06-PRIMARY CON-ROD
2	09E-39-00-2-07-SECONDARY CON-ROD
2	09E-39-00-2-08-SLIDE VALVE+ROD
2	09E-39-00-2-09-SLIDE VALVE GUIDE YOKE
1	09E-39-00-2-10-SLIDE VALVE ROCKING SHAFT
1	09E-39-00-2-10-SLIDE VALVE ROCKING SHAFT
2	09E-39-00-2-11-ECCTERIC STRAP
2	09E-39-00-2-12-SLIDE VALVE ROD LINK
8	09E-39-00-M2.5x4.5 A-K GRUB SCREW
71	09E-39-00-M3 NUT-BRASS
24	09E-39-00-M3 WASHER-BRASS
24	09E-39-00-M3x10 A-K CYL HEAD SCREW
8	09E-39-00-M3x14 A-K C-SINK SCREW
22	09E-39-00-M3x18 A-K CYL HEAD SCREW
4	09E-39-00-M3x3 A-K GRUB SCREW
16	09E-39-00-M3x30 THREADED ROD-BRASS
8	09E-39-00-M3x8 C-SINK SCREW-BRASS
2	09E-39-00-M4 NUT-BRASS
2	09E-39-00-M4 WASHER-BRASS
12	09E-39-00-M4x16 A-K CYL HEAD SCREW
7	09E-39-00-M4x18 A-K C-SINK SCREW
2	09E-39-00-M4x7 A-K GRUB SCREW
6	09E-39-00-M5 NUT
2	09E-39-00-M5x28 A-K C-SINK SCREW
4	09E-39-00-M6 DOME NUT
8	09E-39-00-M6 NUT
2	09E-39-00-M6x40 A-K C-SINK SCREW
2	09E-39-00-M8 NUT

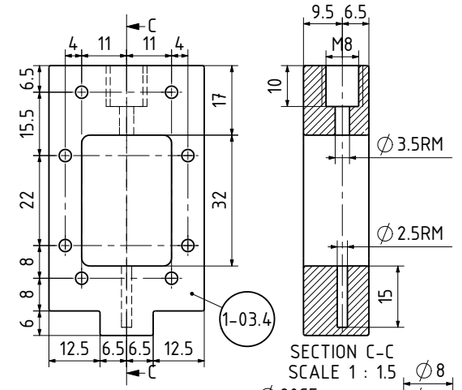
NOTES:
0. ALL DRAWINGS ARE IN METRIC MEASUREMENTS
1. ALL ENGINEERING PRACTICES SHALL BE APPLIED WITH REGARDS TO HOLE AND SHAFT TOLERANCES.
2. WHERE SCREWS OR BOLTS ARE USED THE CLEARANCE HOLES SHALL BE APPROXIMATELY 5% TO 8% LARGER THAN THE MATCHING TAPPED HOLE.
3. PREFERABLY ALL TAPPED HOLES AND MATCHING SCREWS AND/OR BOLTS TO BE METRIC FINE (MF)
4. MATERIALS SPECIFIED ON THE DRAWINGS ARE INDICATIVE ONLY. THE BUILDER CAN MAKE HIS/HER OWN MATERIAL CHOICE.
5. ALL CONNECTIONS/JOINTS WHICH HAVE STEAM PRESSURE APPLIED TO IT SHALL BE SILVER/HARD SOLDERED.
6. COMPRESSION SPRINGS ARE DRAWN IN COMPRESSED STATE (CP), UNCOMPRESSED STATE IS APPROX 40% TO 60% LONGER THEN COMPRESSED STATE.
7. WHERE PREFERRED SCREW OR RIVETED CONNECTIONS CAN BE OMITTED AND PARTS CAN BE BONDED TOGETHER BY USING EITHER HIGH STRENGTH GLUE, EPOXY RESIN, OR SOLDER.
8. PARTS WHICH ARE DIRECTLY EXPOSED TO STEAM AND/OR WATER SHOULD BE CONSTRUCTED USING NON-FERROUS OR NON CORROSIVE MATERIAL SUCH AS BRASS, BRONZE, GUNMETAL, STAINLESS STEEL, COPPER OR MONEL.
9. THE ORDER IN WHICH THE PARTS/COMPONENTS ARE MANUFACTURED AND THE MODEL IS ASSEMBLED IS ENTIRELY LEFT TO THE BUILDER/MODEL MAKER.
10. A COLOUR SCHEME FOR THIS PROJECT IS ENTIRELY LEFT UP TO THE MODEL MAKER.
11. THE MANNER IN WHICH THE PARTS/COMPONENTS ARE MANUFACTURED IS ENTIRELY LEFT UP TO THE BUILDER.
12. USE LOCTITE, ON SCREW OR PRESS FIT CONNECTIONS OR SURFACES, WERE DEEMED NECESSARY TO PREVENT PARTS FROM LOOSENING.
13. WASHERS AND/OR SPRINGWASHERS SHALL BE USED WHERE DEEMED NECESSARY.
XX. ERRORS AND/OR OMISSIONS MAY OCCUR IN THE DRAWINGS, DO NOT HESITATE TO CONTACT ME SO THAT THE ERRORS/OMISSIONS CAN BE RECTIFIED.

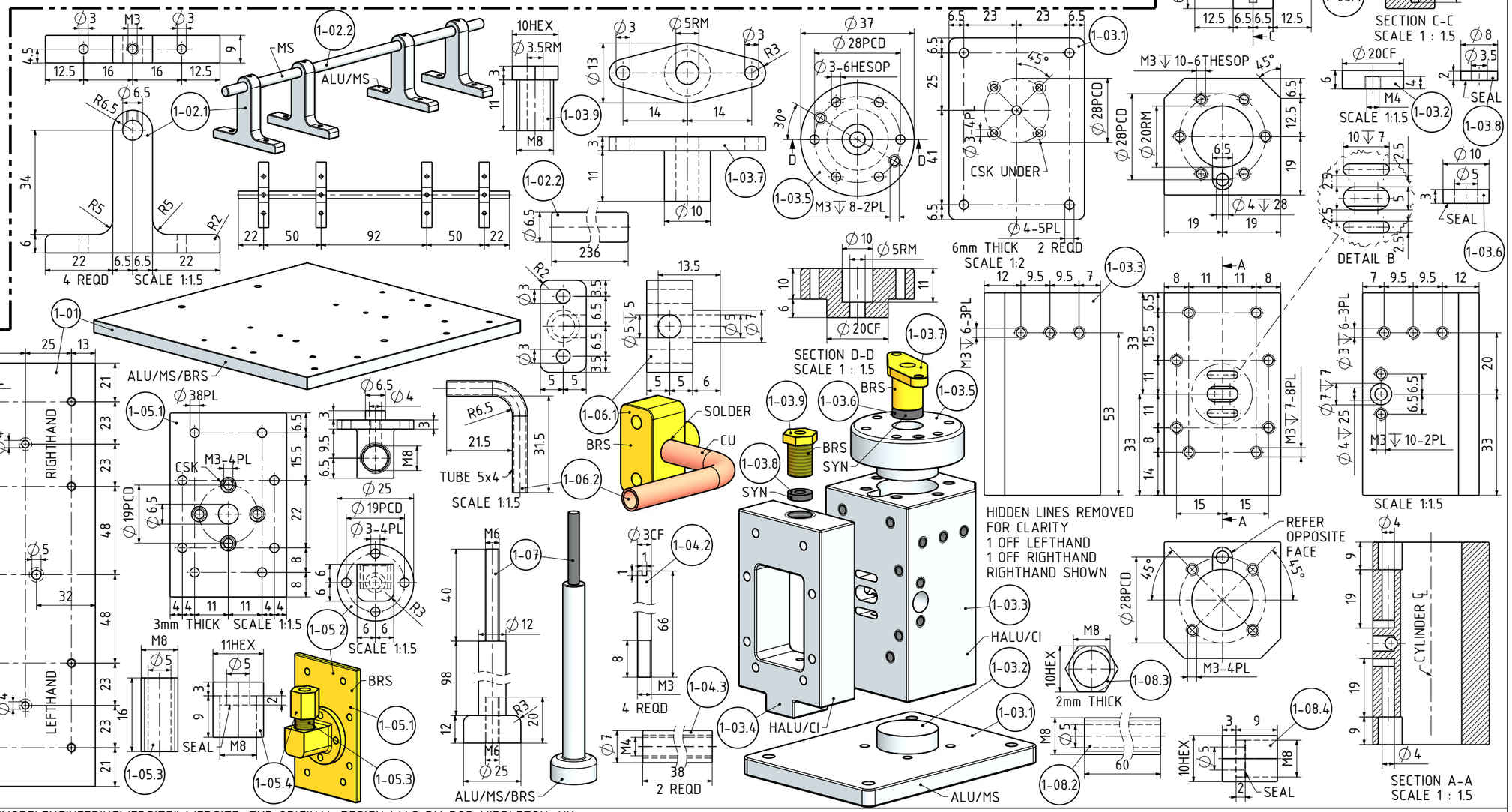
OTHER ABBREVIATIONS
DP = DEEP
DAA= DRILL AFTER ASSEMBLY
D&TAA= DRILL AND TAP AFTER ASSEMBLY
CF = CLOSE FIT (SIZE FOR SIZE)
PF = PRESS FIT
PFAA= PRESS FIT AFTER ASSEMBLY
PCD = PITCH CIRCLE DIAMETER
RM = REAM
HEX = HEXACON, 6SIDED
CP = COMPRESSED
KNL = KNURLED
CSK = COUNTERSINK
PL = PLACES
DWL= DOWEL
SPF= SPOTFACE
(T)HESOP=(TAPPED)HOLES EQUALLY SPACED ON PCD
(T)HESOC=(TAPPED)HOLES EQUALLY SPACED ON CIRCUMFERENCE
OD = OUTSIDE DIAMETER
ID = INSIDE DIAMETER
SA-xxx = SUB ASSEMBLY-xxx

MATERIAL ABBREVIATIONS:
ALU = ALUMINIUM
HALU= HARD ALUMINIUM
BRS = BRASS
BRZ = BRONZE OR GUNMETAL (BRZ/GM)
CI = CAST IRON
CU = COPPER
GRA = GRAPHITE
MS = MILD STEEL/BRIGHT MILD STEEL
SS = SILVER STEEL OR STAINLESS STEEL
SPS = SPRING STEEL
PEEK= POLYETHER ETHER KETONE
SYN = SYNTHETIC MATERIAL SUCH AS VETON, NYLON, TEFLON OR RUBBER
IN GENERAL SYNTHETIC MATERIALS SOULD BE ABLE TO WITHSTAND THE HEAT AND PRESSURE(S) APPLIED TO THEM.
nnn/nnn MEANS THAT EITHER MATERIAL CAN BE USED



THE OFF SET ANGLE OF THE ECCENTRIC IN RELATION TO THE CRANK AXIS TO BE EXPERIMENTALLY DETERMINED FOR THE SMOOTH RUNNING OF THE ENGINE AND SATISFACTION OF THE BUILDER





NOTES: I DOWNLOADED THE ORIGINAL DRAFTS FROM THE "MODELENGINEERINGWEBSITE" WEBSITE. THE ORIGINAL DESIGN WAS BY BOB MIDDLETON, UK.

TITLE
A MODEL OF A 2 CYLINDER DOUBLE SIDED BEAM STEAM ENGINE BY BOB MIDDLETON

DRAWING CONTENTS
NOTES, BILL OF MATERIALS, PARTS AND ASSEMBLIES

PROJECT No 09E-39-00
JDW DRAUGHTING SERVICES
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PROJECTION | JDWDS | MODEL SCALE: 1:1
DATE JANUARY 2020 | DWG SCALE: 1:1 @A3 OR AS SHOWN
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SHEET: 02 OF 05 | A3 | No: 09E-39-00-SHT02

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